Data Title: "FICS-PCB: A Multi-Modal Image Dataset for Automated Printed Circuit Board Visual Inspection"

Abstract
This dataset is designed to support the evaluation of automated PCB visual inspection systems. It collected at the SeCurity and AssuraNce (SCAN) lab at the University of Florida. This dataset consists of 9,912 PCB sample images collected from a DSLR camera and a digital optical microscope, and the annotation results to 77,347 component images. This dataset was used for performance evaluation on state-of-the-art PCB component classification methods. The detailed dataset description is in the reference paper.

We would like to hear your usage of the data, and please feel free to contact us through qslvhw@ufl.edu.

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Details of each zip file

Each zip file contains images from one PCB sample. For example, following files are included after unzipping s1.zip file:

```text
 â€”s1
     |---- DSLR
     |         |---- annotation
     |         |         |---- s1_front.csv
     |         |---- components
     |         |         |---- front
     |         |         |        |---- capacitors/*.png
     |         |         |        |---- ICs/*.png
     |         |         |        |---- inductors/*.png
     |         |         |        |---- resistors/*.png
     |         |---- img
     |         |         |---- s1_front.tif
     |
     |---- Microscope
     |         |---- annotation
     |         |         |---- front/s1_front.csv
     |         |---- components
     |         |         |---- front
     |         |         |        |---- 1.5x (magnification, including 1x, 2x)
     |         |         |        |         |---- capacitors/*.png
     |         |         |        |         |---- ICs/*.png
     |         |         |        |         |---- inductors/*.png
     |         |         |        |         |---- resistors/*.png
     |         |---- img
     |         |         |---- front (PCB side, some PCB have backside image annotated)
     |         |         |        |---- 1.5x (1x, 2x)
     |         |         |        |         |---- s1_p1_1x_20_ring (p1: the number of SEM scan operations. 20: intensity, including 40, 60)
     |         |         |        |         |        |---- TileScan_001/*.tif
     |         |         |        |         |        |---- a automated cropped â€œ.pngâ€ image by SEM software
     |         |         |        |         |---- ...
     |         |         |        |---- ...
```

> ```pyScripts.zip``` includes two python scripts that help to extract annotated components from PCB images.

The size of each zip file is:

s1: 1.3G	s6: 2.67G	s11: 9.97G	s16: 2.76G	s21: 1.17G	s26: 9.81G	s31: 2.39G
s2: 2.85G	s7: 2.77G	s12: 4.55G	s17: 0.85G	s22: 1.37G	s27: 0.6G	
s3: 2.81G	s8: 1.63G	s13: 0.39G*	s18: 0.15G	s23: 1.35G	s28: 0.56G	
s4: 1.25G	s9: 2.82G	s14: 9.04G	s19: 0.67G	s24: 1.42G	s29: 0.56G	
s5: 2.64G	s10: 2.55G	s15: 9.6G	s20: 0.77G	s25: 1.49G	s30: 1.35G	

> NOTE: *s13 does not have DSLR images.

References

- FICS-PCB: A Multi-Modal Image Dataset for Automated Printed Circuit Board Visual Inspection, available at: https://eprint.iacr.org/2020/366.pdf